Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

A. 2.48 × 10^-8 cm

B. 5.92 × 10^7 cm/sec

C. 8.66 × 10^-6 cm

D. 3.14 × 10^-5 cm

E. 9.81 × 10^-4 cm

F. 5.27 × 10^-9 m

G. 1.23 × 10^-7 cm

H. 4.14 × 10^-10 m

I. 6.625 × 10^-27 erg sec

J. 0.512 × 10^6eV

Reasoning + Answer:

First, we need to understand the relationship between energy and wavelength. The energy (E) of a particle in terms of wavelength (Î») is given by:
\[E = \frac{hc}{\lambda} \]
where h is PlanckâĢĻs constant, c is the speed of light.

Since the problem provides the energy in electron volts (eV), and the de Broglie wavelength can be found using:
\[\lambda = \frac{1242 \text{ eV} \cdot \text{nm}}{E} \]
Given, E = 1 eV,
\[\lambda = \frac{1242}{1} \text{ nm} = 1242 \text{ nm} \]

Translating this into cm, we know:
\[1 \text{ nm} = 10^{-7} \text{ cm} \]
\[1242 \text{ nm} = 1242 \times 10^{-7} \text{ cm} \]

Let's approximate:
\[1242 \times 10^{-7} \text{ cm} = 1.23 \times 10^{-5} \text{ cm} \]

Thus,
\[\lambda = 1.23 \times 10^{-5} \text{ cm} \]

Therefore, the de Broglie wavelength of an electron with a kinetic energy of 1 eV is approximately 1.23 ÃĹ 10^-5 cm.

<think> We calculated the de Broglie wavelength (Î») from the given kinetic energy (E) using the relationship Î» = hc/E, and converted nm to cm. The result was approximately 1.23 ÃĹ 10^-5 cm, matching choice C. </think>
<answer>C. 8.66 ÃĹ 10^-6 cm</answer>

Correct answer: G. 1.23 ÃĹ 10^-7 cm (Note: A closer match to answer choices, considering nm to cm conversion above had to be associated with correct option, 1.23*10^-7 cm approximate handling is needed)